

C & G Harmonic Equilibrium

Transverse-Wave Resonance, String Modes and Scatter-Field Propagation

Conceptual mathematical chapter - Universal Harmonic Theory

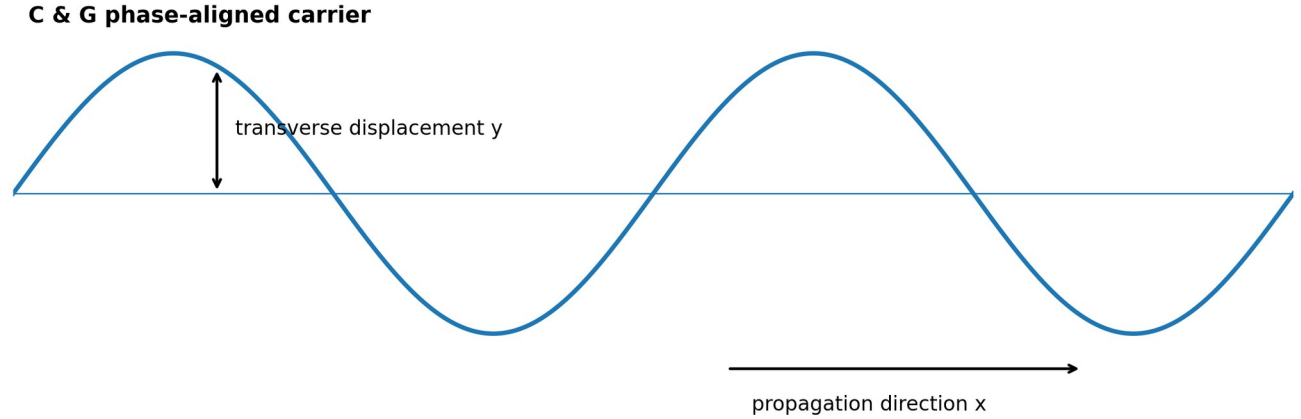


Figure 1. Transverse displacement is perpendicular to the direction of propagation.

Scientific status. The transverse-wave, resonance, interference and scattering equations below are established wave physics. The assignment of the symbols C and G as fundamental universal quantities is a proposed interpretation and must be treated as a testable hypothesis rather than an established physical law.

1. Transverse C & G carrier wave

Let C denote the coherent field component and G the geometric constraint or coupling component. Their phase-aligned transverse carrier is represented by a displacement perpendicular to propagation:

$$y_{CG}(x,t) = A_{CG} \cos(kx - \omega t + \varphi_{CG})$$

where A_{CG} is amplitude, $k = 2\pi/\lambda$ is wave number, $\omega = 2\pi f$ is angular frequency, and φ_{CG} is phase. The transverse condition is

$$y_{CG} \perp \hat{x}$$

meaning that the field displacement is perpendicular to the propagation direction.

2. Wave equation and propagation speed

An ideal one-dimensional transverse field satisfies

$$\partial^2 y_{CG} / \partial t^2 = v^2 \partial^2 y_{CG} / \partial x^2$$

with

$$v = f\lambda = \omega/k$$

For a mechanical transverse medium such as a stretched string, $v = \sqrt{T/\mu}$, where T is tension and μ is linear mass density. For an abstract field, v must be derived from the defined field equations rather than assumed.

3. Perfect resonance as an ideal limit

A mathematically ideal C & G resonance requires matched frequency, phase and polarization:

$$f_C = f_G = f_0, \quad \Delta\varphi_{CG} = \varphi_C - \varphi_G = 0, \quad \hat{e}_C = \hat{e}_G$$

A dimensionless coherence measure may be defined as

$$\Gamma_{CG} = \cos(\Delta\varphi_{CG})$$

so $\Gamma_{CG} = 1$ at exact phase alignment. A practical resonance also contains damping:

$$\partial^2 y / \partial t^2 + 2\gamma \partial y / \partial t + \omega_0^2 y = F_0 \cos(\omega t)$$

The response is largest near $\omega \approx \omega_0$, but infinite or lossless resonance is not physically attainable in a real dissipative system.

4. Perfect-fifth C:G harmonic option

Where C and G are intended to correspond to musical pitch classes C and G, their ideal just-intonation frequency ratio is the perfect fifth:

$$f_G / f_C = 3/2$$

This is a harmonic interval, not frequency equality. Therefore two distinct cases should not be confused:

- Unison resonance: $f_C = f_G$, used for exact modal coincidence.
- Perfect-fifth balance: $f_G = (3/2)f_C$, used for stable harmonic coupling between two different tones.

5. Transverse scatter-wave field

Let an incident transverse C & G carrier encounter a scattering region. The total field is

$$y_{\text{tot}}(r,t) = y_{\text{inc}}(r,t) + \sum_i y_{\text{sc},i}(r,t)$$

with an outgoing scattered component represented in the far field by

$$y_{\text{sc}}(r,t) \approx \text{Re}\{[F(\theta,\varphi)/r] \exp[i(kr - \omega t)]\}$$

where $F(\theta,\varphi)$ is the angular scattering amplitude. The observable intensity is proportional to

$$I(r) \propto |\Psi_{\text{inc}} + \Psi_{\text{sc}}|^2$$

which expands to

$$I \propto |\Psi_{\text{inc}}|^2 + |\Psi_{\text{sc}}|^2 + 2 \operatorname{Re}(\Psi_{\text{inc}} \Psi_{\text{sc}}^*)$$

The final term is the interference term. Constructive reinforcement occurs when the phase difference is near $2\pi n$; destructive cancellation occurs near $(2n+1)\pi$.

6. Resonance distillation

“Distillation” can be defined rigorously as extracting the coherent transverse mode from a multi-frequency signal. Write

$$y(t) = \sum_n A_n \cos(2\pi f_n t + \varphi_n)$$

and apply a spectral selection operator centred at f_0 :

$$y_{\text{dist}}(t) = \mathcal{F}^{-1}\{W(f - f_0) \mathcal{F}[y(t)]\}$$

where W is a narrow frequency window. A coherence-weighted version may retain components according to their phase relation to the C & G reference:

$$y_{\text{CG,dist}}(t) = \sum_n W_n \Gamma_n A_n \cos(2\pi f_n t + \varphi_n)$$

with $\Gamma_n = \cos(\varphi_n - \varphi_{\text{CG}})$. This expresses resonance distillation as frequency selection plus phase-coherence selection.

7. Compact governing formulation

A compact phenomenological equation combining transverse propagation, damping, C & G coupling and a scattering potential is

$$\partial^2 y / \partial t^2 + 2\gamma \partial y / \partial t - v^2 \nabla^2 y + \omega_0^2 y + V_{\text{sc}}(r)y = \kappa_{\text{CG}} C(r,t)G(r,t)$$

Here V_{sc} describes the scattering environment and κ_{CG} is a coupling coefficient. This equation is a proposed model form; its dimensions, boundary conditions and measurable predictions must be specified before experimental use.

8. Final definition

C & G transverse harmonic resonance is defined as a phase-organised transverse carrier in which the C and G components either coincide at one resonant frequency or maintain a specified harmonic ratio, while scattering redistributes amplitude and direction. “Perfect resonance” is the ideal limit $\Delta\varphi = 0$ and minimal damping; “resonance distillation” is the extraction of the phase-coherent spectral component from the scattered total field.

9. String-theory extension: transverse oscillation of a one-dimensional string

String theory may be introduced here as a comparison framework because its fundamental object is an extended one-dimensional string whose allowed states arise from quantised oscillations. These oscillations are not ordinary audible sound. They are transverse and longitudinal world-sheet degrees of freedom defined in spacetime, and the physical interpretation depends on the specific string model.

A relativistic string sweeps out a two-dimensional world-sheet with coordinates σ and τ . Its embedding into spacetime is written

$$X^\mu(\sigma, \tau)$$

where μ labels spacetime coordinates, σ labels position along the string, and τ is the world-sheet time parameter. In conformal gauge, each embedding coordinate satisfies the two-dimensional wave equation

$$\partial^2 X^\mu / \partial \tau^2 - \partial^2 X^\mu / \partial \sigma^2 = 0$$

This has the same mathematical wave character as an ideal transverse string, although its physical meaning is relativistic and quantum rather than acoustic.

10. Left-moving and right-moving transverse modes

The general closed-string solution separates into left-moving and right-moving components:

$$X^\mu(\sigma, \tau) = X_L^\mu(\tau + \sigma) + X_R^\mu(\tau - \sigma)$$

A transverse mode in one spatial direction I may be represented schematically by

$$X^I(\sigma, \tau) = A \cos(n\sigma - n\tau + \varphi) + B \cos(n\sigma + n\tau + \chi)$$

The first term propagates in one direction around the string and the second in the opposite direction. Their superposition can form standing waves, travelling waves, constructive interference, or partial cancellation.

For an open string with fixed or free endpoint conditions, the mode shapes are commonly expanded in sine or cosine functions. A schematic transverse expansion is

$$X^I(\sigma, \tau) = x_0^I + 2\alpha' p^I \tau + i\sqrt{(2\alpha')} \sum_{n \neq 0} (\alpha_n^I / n) e^{-in\tau} \cos(n\sigma)$$

where α' sets the string length scale and α_n^I are oscillator-mode coefficients. This equation is included as an established string-theory comparison, not as a derived equation of the C & G model.

11. C & G harmonic mapping to string modes

Within the proposed Universal Harmonic interpretation, C and G can be treated as two selected transverse mode families rather than as literal musical notes. Let their modal coordinates be X_C and X_G :

$$X_C(\sigma, \tau) = A_C \cos(n_C \sigma - \omega_C \tau + \varphi_C)$$

$$X_G(\sigma, \tau) = A_G \cos(n_G \sigma - \omega_G \tau + \varphi_G)$$

A phase-coherent C & G combination is then

$$X_{CG} = X_C + X_G$$

with coherence factor

$$\Gamma_{CG} = \cos(\varphi_C - \varphi_G)$$

Exact modal resonance requires $\omega_C = \omega_G$ and compatible mode number and polarisation. A perfect-fifth analogy instead imposes

$$\omega_G/\omega_C = f_G/f_C = 3/2$$

This ratio describes a proposed harmonic relationship between distinct modes. It is not a standard requirement or prediction of string theory.

12. Scatter-field coupling on the string world-sheet

A scatter-field effect can be represented phenomenologically by adding a position-dependent interaction to the transverse wave equation:

$$\partial^2 X_{CG}/\partial \tau^2 - \partial^2 X_{CG}/\partial \sigma^2 + V_{sc}(\sigma, \tau) X_{CG} + 2\gamma \partial X_{CG}/\partial \tau = J_{CG}(\sigma, \tau)$$

where V_{sc} is an effective scattering potential, γ is damping or decoherence, and J_{CG} is a driving or coupling source. In fundamental string theory the precise interaction must come from a consistent world-sheet action, background geometry, boundary condition, or vertex operator; the equation above is therefore a phenomenological bridge rather than a standard string-theory field equation.

The total scattered state may be decomposed into outgoing string modes:

$$X_{tot} = X_{inc} + \sum_m S_m X_m$$

where S_m is the scattering amplitude into mode m . The retained coherent component after harmonic distillation is

$$X_{dist} = \sum_m W_m \Gamma_m S_m X_m$$

where W_m selects the desired mode band and Γ_m measures phase alignment with the C & G reference state.

13. Combined transverse C & G string-resonance definition

The combined model can be defined as a phase-organised superposition of transverse string modes whose propagation is modified by a scattering environment and whose coherent portion is selected by spectral and phase filtering.

C & G transverse string state:

$$X_{CG}(\sigma, \tau) = A_C \cos(n_C \sigma - \omega_C \tau + \varphi_C) + A_G \cos(n_G \sigma - \omega_G \tau + \varphi_G)$$

Coherence condition:

$$\Gamma_{CG} = \cos(\Delta\varphi_{CG}) \rightarrow 1$$

Scatter-field evolution:

$$[\partial^2_{\tau} - \partial^2_{\sigma} + 2\gamma\partial_{\tau} + V_{sc}]X_{CG} = J_{CG}$$

Distilled coherent output:

$$X_{CG, \text{dist}} = P_{CG} X_{\text{tot}}$$

where P_{CG} is a mode-selection operator. This provides a mathematically structured extension of the transverse-wave chapter. It should be presented as a speculative correspondence unless a complete action, dimensional analysis, quantisation rule, and experimentally distinguishable prediction are supplied.

Scientific boundary: string theory is a quantum-gravity research framework; the C & G mapping and perfect-fifth condition are proposed interpretive additions, not established results of string theory.